

Ahmad Mahmood

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Education

ETH Zurich

MSc Computer Science | GPA: 5.4 / 6

Sep 2023 - Jan 2026

Zurich, Switzerland

Lahore University of Management Sciences (LUMS)

BSc Computer Science | GPA: 3.8 / 4

Sep 2019 - May 2023

Lahore, Pakistan

Experience

Cast Insights

Machine Learning Engineer

Jul 2025 - Present

San Francisco, USA

- Built an **agentic pipeline** using **Pydantic AI** with tools including a PCA-subspace RAG pipeline projecting query embeddings onto the corpus manifold, achieving **NDCG@1 +0.44%**, **Recall@1 +0.39%** improvement on Natural Questions.
- Built and shipped a **story detection** and **recommendation system** using NER and POS tagging, improving story quality score by **46%**.
- Deployed HNSW index on text embeddings for **91% similarity search speedup**.
- Implemented and maintained an **MCP server** using the **FastMCP** SDK and OAuth 2.0 authentication flow for client integrations.

Disney

Machine Learning Engineer

Jan 2025 - Sep 2025

Zurich, Switzerland

- Implemented a diffusion-SSM hybrid model for image deblurring by integrating diffusion generative priors with Mamba state-space sequence models.
- Built a custom hidden-state initialization for Mamba selective scan kernels that injects diffusion priors at **2x overhead vs. 10x** for the naive approach, without modifying CUDA kernels.

INSAT

Computer Vision Engineer (Research Intern)

Mar 2024 - Oct 2024

Sofia, Bulgaria

- Built InTraGen, a trajectory-conditioned diffusion transformer (DiT) video generation pipeline achieving **MTEM 9.40% vs. 22.17%** baseline and **FID 26.37 vs. 40.65**; outperformed LUMA, I2VGen-XL, SEINE. Accepted to **ICCV 2025**.
- Designed and implemented MTEM, a trajectory evaluation metric using bipartite graph matching (Hungarian algorithm); adopted as the primary benchmark for the paper.
- Engineered the ViN dataset pipeline - **50K synthetic videos** across 4 physics-simulated interaction subsets using Blender and Unity.

Mohamed bin Zayed University of AI

Research Engineer (Intern)

Mar 2022 - Dec 2022

Abu Dhabi, UAE

- Implemented temporal prompt optimization on frozen vision models (ViTs, DINO, CLIP) to boost cross-modal adversarial transferability (**ICLR 2023**); built VURF, an LLM-based video understanding pipeline for QA and reasoning (**NeurIPS 2024**).

Projects

3D Scene Reconstruction from Multi-View Images | PyTorch, Open3D

2025

- Built a 3D reconstruction pipeline from multi-view images using differentiable rendering, camera pose estimation, feature matching, and volumetric rendering for novel view synthesis.

Conditional Diffusion Models for Image Inpainting | PyTorch, DDPM

Jun 2025

- Implemented a class-conditional diffusion model (multi-scale UNet) from scratch; applied to inpainting by conditioning the reverse diffusion process on partially observed pixels.

Neural Network Robustness Verification (DeepPoly)

2024

- Implemented a neural network verifier using DeepPoly convex relaxation to certify adversarial robustness on MNIST and CIFAR-10; analyzed tightness vs. computational cost trade-offs.

Selected Publications

Boosting Adversarial Transferability using Dynamic Cues - Mahmood et al., **ICLR 2023**

InTraGen: Trajectory-controlled Video Generation for Object Interactions - Mahmood et al.

VURF: General-purpose Reasoning and Self-refinement Framework for Video Understanding - Mahmood et al., **NeurIPS 2024 Workshop**

Honors and Awards

International Mathematics Olympiad (IMO) 2019 Top 6 in Pakistan (Bath, UK) **Dean's Honour List** Years 1-3 at LUMS (cGPA **3.85-3.93/4.0**)

Google Kickstart 1st place (Pakistan), multiple rounds

Technical Skills

Languages: Python, C/C++, JavaScript, Haskell, LaTeX

Frameworks: PyTorch, Diffusion Models (DDPM/DiT/Score-based), LangChain, LangGraph, Flask, ReactJS, FastMCP

Infrastructure: AWS, Google Cloud, CUDA, Git, Docker